

Week 1: Daily Journals

Monday:

Today marked the commencement of the 2026 Summer Research Program. My group began with a kickoff meeting to review the project overview and objectives for the six-week program. Following the meeting, I conducted an inventory check to identify the necessary components for fabricating four distinct cables required for our upcoming testing. Lastly, I reviewed our Python-based Mode Manager program to verify that it functions as predicted ahead of our Flatsat testing.

Tuesday:

On the second day, my group began fabricating the required cabling. We initiated the process by soldering six 22-gauge wires into a 9-pin D-sub male connector; this cable will handle power distribution between our three power supplies and the switchboard. Next, we began soldering 25 individual 24-gauge wires to a 25-pin D-sub male connector, which will interface the switchboard with the Raspberry Pi 4's GPIO pins. To optimize efficiency, we established a two-person pipeline: one person handled the soldering while the other prepared the subsequent wires and applied labels. Our specific pinout configuration is documented in Figure 1.

Wednesday:

Today, we completed fabrication for the majority of the remaining cables. This included assembling a 9-pin D-sub male connector cable to interface the Switchboard with the Antenna Interface Board (AIB), as well as another 9-pin D-sub male connector cable intended for the Switchboard-to-camera interface. Once assembly was complete, I organized and laid out all available components and cabling for the Flatsat integration (displayed in Figure 2). We are currently awaiting the delivery of the switchboard, Pi Hat, X-Ray sensor, and the dedicated X-Ray sensor cable.

Thursday:

Today, we identified a design discrepancy: the cable interfacing the switchboard and the cameras required a 15-pin D-sub connector rather than the 9-pin connector we had assembled. To correct this, we desoldered the 9-pin connector and resoldered the wire harness to a 15-pin connector. Afterward, I began developing a Python-based diagnostic utility to verify that the Pi 4 software executes correctly within the Flatsat environment. This script is designed to run prior to testing cycles to ensure hardware health. The program executes the following sequential routine:

1. Runs a basic hardware diagnostic test on the Pi 4.
2. Monitors the ESim heartbeat signal.
3. Verifies Ethernet-over-USB commanding functionality from the ESim to the Pi.
4. Confirms successful data file retrieval from the ESim.
5. Verifies commanded mode transitions and ensures a notification packet is transmitted back to the ESim upon every transition.

Friday:

Today was dedicated to refining and finalizing the Pi 4 software diagnostic program. I focused on optimizing the code structure to ensure all test cases execute reliably. I plan to run full integration testing with this script on Monday. With development wrapped up for the day, I concluded my tasks for the week.

Pictures

Power In Pinout	GPIO In Pinout	X-ray Sensor Pinout	Camera and Pi4 Pinout
1 +5V	1 GPIO 23 +5V Switch (AIB/SDR 1)	1 GND	1 +5V Camera
2	2 GPIO 25 +3.3V Switch (AIB)	2 +5V	2 Camera Select
3 +3.3V	3 GPIO 27 +12V Switch (AIB)	3 SPI SCLK	3
4	4 GND (Pin 6 on GPIO Header)	4 SPI MOSI	4
5 +12V	5 GPIO 14 UART TX (AIB)	5 SPI MISO	5 GND
6 GND	6 GPIO 15 UART RX (AIB)	6 SPI CEO	6 GND
7	7 GPIO 22 +5V Switch (X-ray)	7	7
8 GND	8 GND (Pin 9 on GPIO Header)	8	8
9 GND	9 GPIO 10 SPI MOSI (X-ray)	9	9 +5V Pi4
The other end connects to the power supplies.	10 GPIO 09 SPI MISO (X-ray)	Wait for CRHS cable.	The other end is open leads.
	11 GPIO 11 SPI SCLK (X-ray)		
	12 GND (Pin 14 on GPIO Header), Twists with pin 11.		
	13 GPIO 08 SPI CEO (X-ray)		AIB Pinout
	14 GPIO 05 +5V Switch (Camera)		1 +5V
	15 GPIO 06 Camera Selection		2
	16 GPIO 26 Pi4 Heartbeat Prompt Output to Pi4		3
	17 GND (Pin 25 on GPIO Header)		4
	18 GPIO 24 PI Heartbeat (PI) Input to Watchdog		5
	19 GND (Pin 30 on GPIO Header)		6
	20 GPIO 12 +5V Switch (Motor)		7 UART Tx
	21 GPIO 13 +5V Switch (Launch Latch)		8 UART Rx
	22 GND (Pin 34 on GPIO Header)		9 GND
	23 GND (Pin 20 on GPIO Header)		The other end is open leads.
	24 GND (Pin 39 on GPIO Header)		
	25 Shield (Case to D-sub Metal Shell)		

Figure 1: Pinout Diagram

